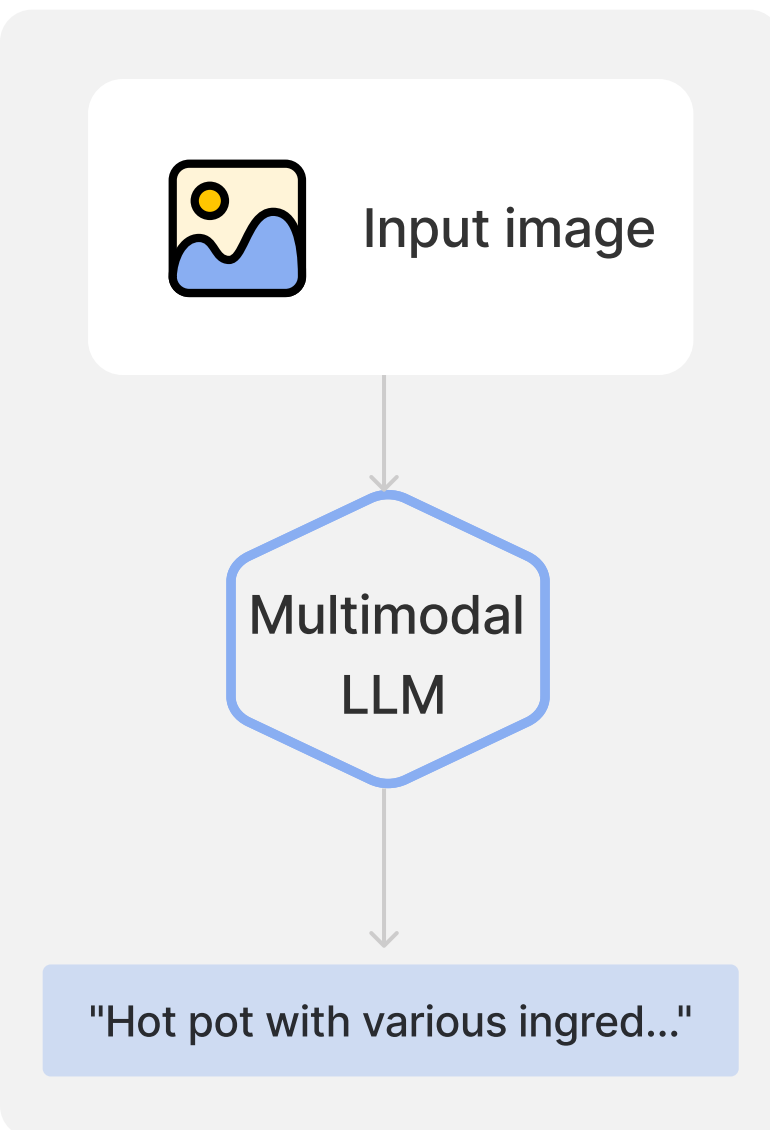
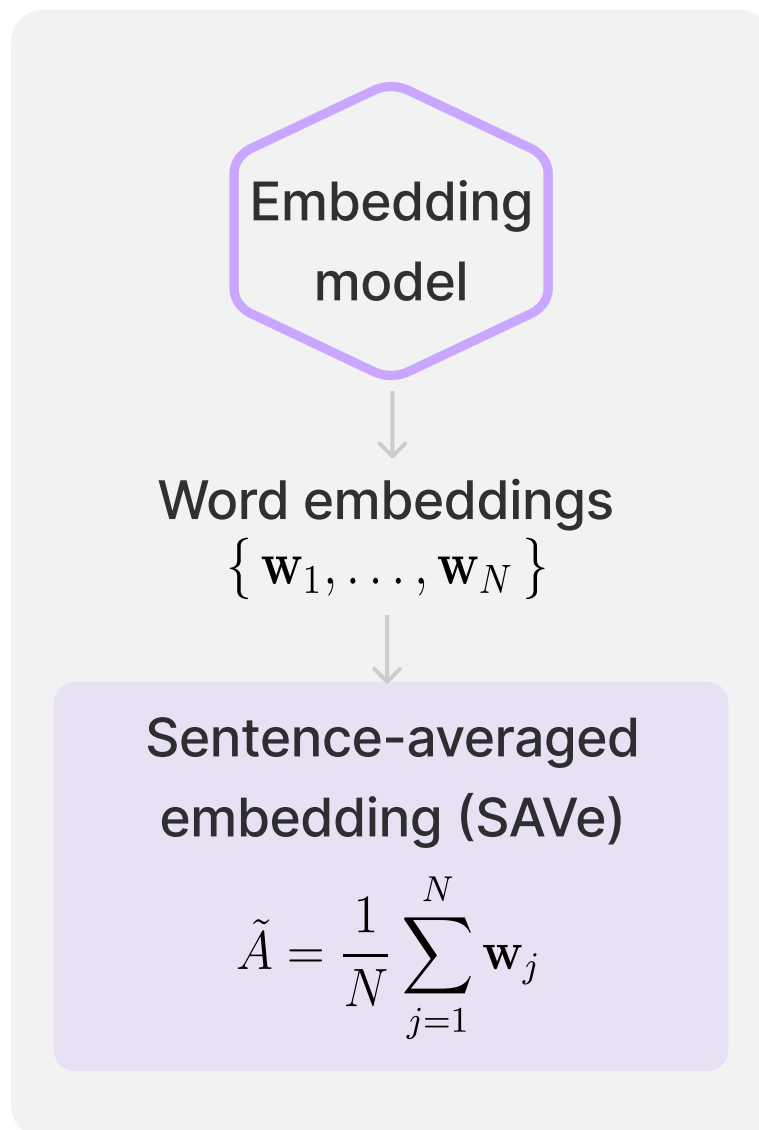


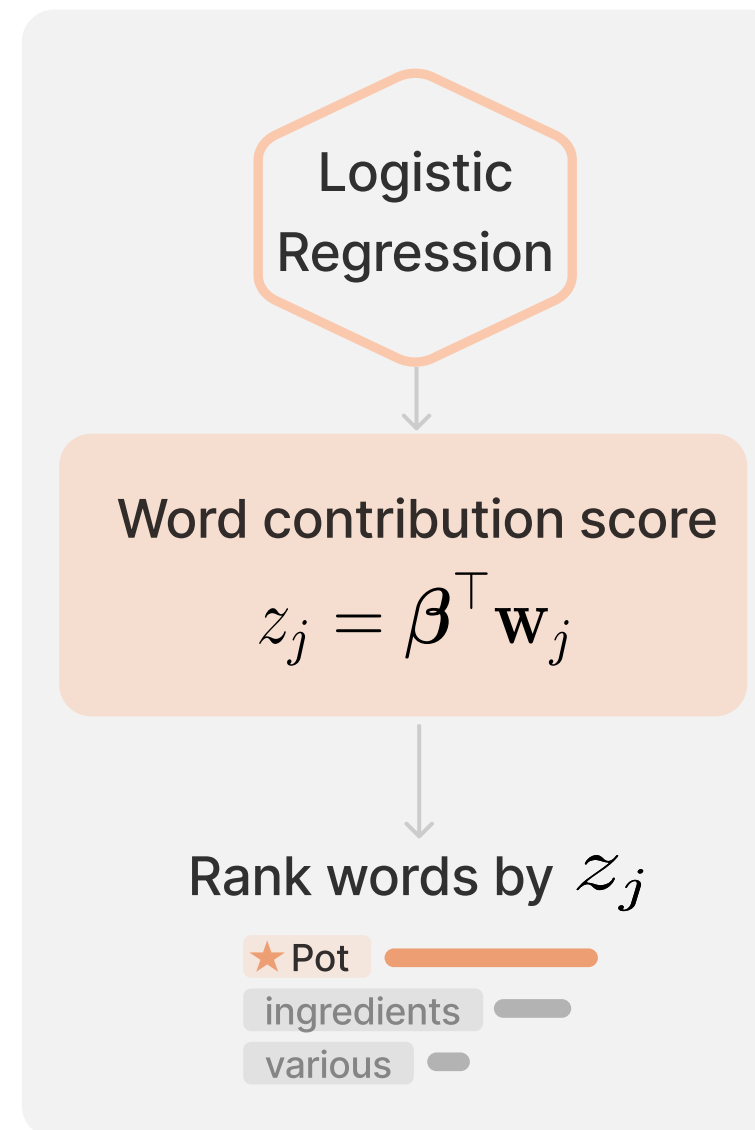
PHASE A — Captioning



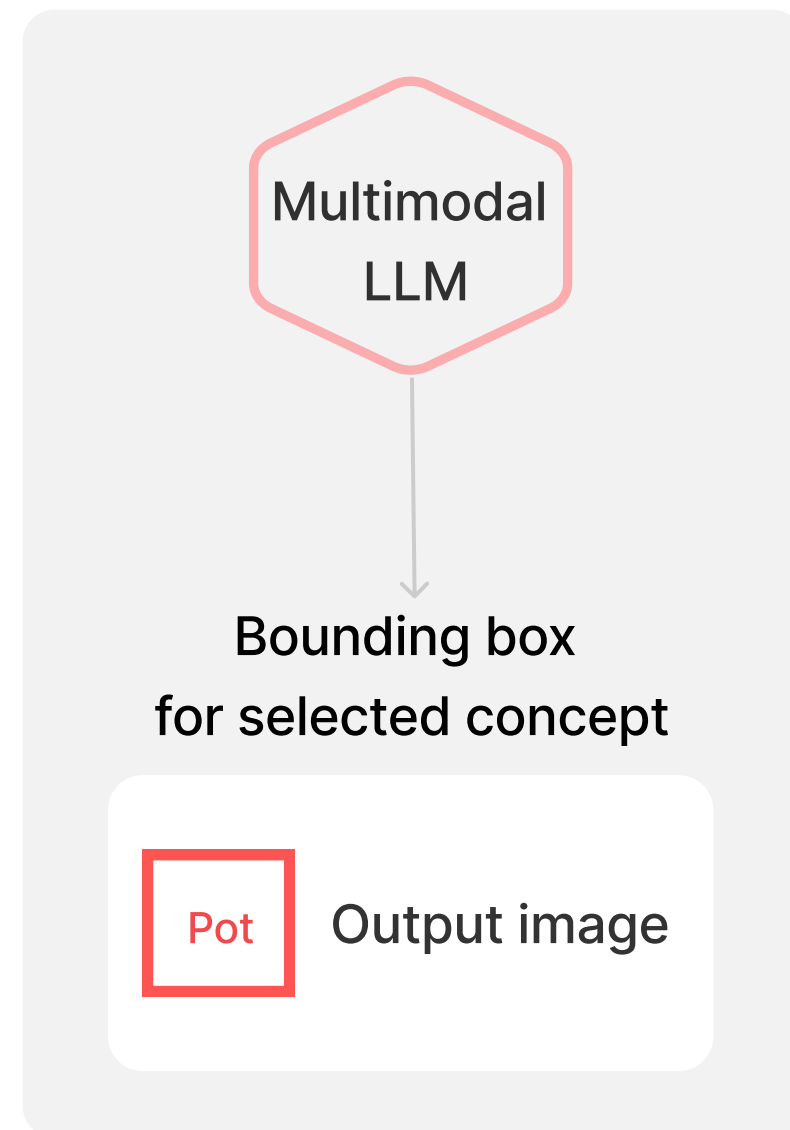
PHASE B — Embedding



PHASE C — Concept discovery



PHASE D — Localization



Notation:

$\mathbf{w}_j$: word embedding

β : classifier weights

N : number of words